

PERSONAL STATEMENT

Jianming Xing

Nanyang Technological University, Singapore
MSc in Signal Processing and Machine Learning

I want to pursue NTU's MSc in Signal Processing and Machine Learning because research made me see robot trajectories as structured signals before training data. As a Research Assistant at iLearn, HIT Shenzhen, I am second author of Hermite Action Tokenizer for VLA, an unsubmitted working paper, and responsible for its discrete autoregressive branch. I separated seven-dimensional action chunks into six-dimensional end-effector trajectories and gripper states, using shared breakpoints to reduce redundancy and quantization seams while coding gripper changes separately. Comparing adaptive MLP-based and fixed Hermite segmentation taught me to ask what temporal structure a representation preserves and to distinguish offline codec fidelity from rollout or hardware performance.

For the August 2027 intake, subject to course availability, EE6401 Advanced Digital Signal Processing would deepen how I model and evaluate trajectory signals rather than treating preprocessing as neutral. EE6222 Machine Vision would strengthen my treatment of visual observations paired with actions. EE6406 Analytic & Ensemble Machine Learning would broaden the methods I can compare and help me reason about what learners retain from a representation. This connects signal structure, perception, and model choice.

If I choose the coursework-only option and EE6008 Collaborative Research & Development Project is available, a team could conduct a bounded study of representation choices. I would hold a dataset and protocol fixed, vary temporal alignment or encoding, and compare how models respond. I would bring experience maintaining clear boundaries among codec metrics, policy evaluation, and physical deployment while learning to organise prototyping, testing, and communication. My goal is AI/ML or learning-systems R&D for embodied agents. NTU would help me build representations whose results remain meaningful when connected to sensing and action.